

Supplementary Materials for Prior-guided image analysis for Bayesian longitudinal maize height phenotyping with fixed cameras

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This file contains 25 pages, three figures, and six tables.

This document provides the detailed implementation specification, causal information-set definition, calibration derivation, uncertainty calibration, candidate-selection audit, annotation provenance, and plant-level endpoint results supporting the longitudinal analysis in the main article. It contains no separate reference list; citations and bibliographic details are given in the main article.

S1 Detailed implementation specification

S1.1 What is and is not Bayesian in the field implementation

Two sequential operations must be distinguished. The field association code maintains a running horizontal position and base row for each plant. It uses a fixed update gain of 0.5 and does not propagate a position covariance, calculate a data-dependent Kalman gain, or retain position particles.

15 This is a fixed-gain state update, algebraically equivalent to a steady-state Kalman-mean recursion
16 after the gain has been specified. It is not a full Bayesian posterior over position.

17 The height and growth component is Bayesian. A weighted particle approximation represents
18 the joint distribution of latent height and growth rate. Particles are propagated through the transition
19 model, weighted with the inlier/outlier measurement likelihood, normalized, and resampled when
20 the effective sample size crosses its prespecified threshold. Posterior summaries and one-step
21 predictive distributions are calculated from those weighted particles.

22 The term *prior-guided image analysis* refers to feedback across the image/model boundary, not
23 simply to the presence of priors in a Bayesian model. The stronger mechanism in which the predicted
24 height distribution re-ranks competing top-keypoint candidates is evaluated in the controlled
25 candidate-ambiguity simulation and in a matched real-image audit. It was not the top-keypoint rule
26 used to produce the primary field-height results. Thus the field implementation supports claims
27 about prior-guided plant association and base-keypoint updating, while the simulation establishes
28 that a height prior can improve top-candidate choice under known ambiguity. The real-image audit
29 tests whether the available natural candidate sets actually contain that ambiguity.

Table S1: Full parameter and implementation specification for the two empirical analysis streams. A dash indicates that the operation was not applied in that stream.

Component	Quantity or operation	2024–2025 daily pole stream	2021 independent manual-reference validation
Observation unit	Plant identity and chronological key	camera_ genotype, left-to-right plant track, capture time	rowid plus plantid, field date
Observation unit	Measurement entering height filter	Depth-corrected red-band pole calibration, cm	Archived single-image mask extent height_sam, multiplied by the recorded geometric scale
Reference	In-scene physical definition	Red-band pole; adjacent band edges 1 ft (30.48 cm) apart	Camera-support pole annotation is an installation asset, not the source of height_sam

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Table S1 continued

Component	Quantity or operation	2024–2025 daily pole stream	2021 independent manual-reference validation
Reference	Depth factor	$D_{\text{tgt}}/D_{\text{ref}} =$ $8.5/10.0 = 0.85$	Separate 2021 scale, 0.415886 cm/pixel, zero intercept
Calibration history	Reference selection	Best passing calibration with timestamp no later than the current image; fit RMSE capped at 0.5 in	Not applicable to archived <code>height_sam</code>
Field association	Expected plants	6 per initialized camera view	Archived identities supplied
Field association	Initialization	First complete six-detection frame, sorted left to right	–
Field association	Assignment	Minimum total absolute horizontal distance; maximum normalized distance 0.16	–

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Table S1 continued

Component	Quantity or operation	2024–2025 daily pole stream	2021 independent manual-reference validation
Field association	Position/base update	$m_t =$ $(1 - K)m_{t-1} + Kx_t$, fixed $K = 0.5$; no covariance state	–
Field association	Empirical diagnostic	2018 matches, 48 more than frozen first-frame anchors	–
Particle filter	Latent state	Height h_t and growth rate r_t	Same
Particle filter	Transition order	Propagate h_t using r_{t-1} , then update r_t	Same
Particle filter	Number of particles	5000	5000
Particle filter	Base random seed	20260723; deterministic plant-specific seed from SHA-256 of plant label and base seed	20260724; same deterministic derivation
Particle filter	Initial height	$N(Y_1, \sigma_1^2)$, clipped to $[0, 450]$ cm	Same

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Table S1 continued

Component	Quantity or operation	2024–2025 daily pole stream	2021 independent manual-reference validation
Particle filter	Initial growth	$N(3, 2^2)$ cm/day, clipped	$N(0, 8^2)$ cm/day, clipped
Particle filter	Height limits	0–450 cm	0–450 cm
Particle filter	Growth limits	-4–15 cm/day	-4–15 cm/day
Particle filter	Height-process SD	16.866 cm/ $\sqrt{\text{day}}$, estimated from depth-corrected 2024 daily increments	8.78 cm/ $\sqrt{\text{day}}$, robust scale of 2021 image increments
Particle filter	Growth-process SD	0.35 cm/day/ $\sqrt{\text{day}}$	0.35 cm/day/ $\sqrt{\text{day}}$
Particle filter	Growth-rate evolution	Random walk ($\rho = 1$)	Nonparametric rate-versus-height field from 2024–2025 pole-stream increments; field adopted before growth noise is added
Particle filter	Outlier probability π	0.0512	0.065
Particle filter	Outlier SD	70.62 cm	59.04 cm

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Table S1 continued

Component	Quantity or operation	2024–2025 daily pole stream	2021 independent manual-reference validation
Particle filter	Resampling rule	Systematic resampling when $ESS < 0.5M$	Same
Particle filter	Posterior interval	Weighted 0.025 and 0.975 height-particle quantiles after the current image	Same
Particle filter	Predictive interval	Numerical 0.025 and 0.975 quantiles of the complete particlewise Gaussian mixture	Same algorithm; field-height comparison emphasizes the posterior interval

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Table S1 continued

Component	Quantity or operation	2024–2025 daily pole stream	2021 independent manual-reference validation
Measurement SD	Inlier scale	Prespecified root- sum-of-squares budget: 8-cm pose term, pole-fit term, 1.5 cm per band outside fitted range, and band-regularity term; the total is multiplied by 0.85 after depth correction	$12[1 + 1.5g_t]$ cm, where g_t is the observed gust normalized over the five evaluated dates; range 12–30 cm, fallback 15 cm
Measurement SD	Interpretation	Pilot uncertainty model; process and measurement variation are not separately identified	Retrospective full-window weather normalization; prospective use requires externally frozen normalization limits

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Table S1 continued

Component	Quantity or operation	2024–2025 daily pole stream	2021 independent manual-reference validation
Primary comparison	Empirical sample	Next image-derived measurements after each first observation	All 132 records, 33 plants, 12 rows, and 5 dates
Primary comparison	Cluster uncertainty	Cameras for calibration summaries	20,000 row-bootstrap replicates; seed 20260908 (endpoint seed 20260909)

S2 Causal online definition

Let I_t be the image acquired at time t , and let $\mathcal{I}_{\leq t} = \{I_1, \dots, I_t\}$ denote the accumulated image pool. With detector weights, thresholds, reference definitions, geometry, and filter settings fixed, the operational update is

$$O_t = A(I_t, O_{t-1}; \theta_{\text{fixed}}), \quad O_t \text{ is measurable with respect to } \mathcal{I}_{\leq t}. \quad (\text{S1})$$

Here O_t contains the association state, physical image measurement, and height-growth particle distribution returned immediately after I_t is added. An image I_{t+k} , $k > 0$, is absent from Equation S1. When it arrives, it initiates a new update; it does not revise O_t .

The field association step predicts each track by carrying forward its running horizontal position

m_{t-1}^x and base row m_{t-1}^b . Current detections are assigned to these carried states by minimum-cost matching. For an accepted observation (x_t, b_t) ,

$$m_t^x = (1 - K)m_{t-1}^x + Kx_t, \quad (S2)$$

$$m_t^b = (1 - K)m_{t-1}^b + Kb_t, \quad K = 0.5. \quad (S3)$$

Equation S3 stores means only. The code does not store a position covariance, so m_t^x and m_t^b are called running position states rather than position posteriors in this supplement.

The physical reference attached to a plant image is also selected as of t : only accepted reference fits whose calibration time is no later than the plant-image capture time are eligible. The selected top and base pixels are converted to centimeters, the depth factor is applied, and the resulting measurement updates the height-growth particle distribution. The cached detection-to-measurement audit recomputed 391 camera-time prefixes and compared 51,035 previously emitted rows, with 0 mismatches.

Primary manual-reference validation uses 2021 plants from 12 stationary-camera rows. These subjects are absent from 2024–2025 development data, and no manual height enters the image measurement, growth field, filter, noise parameters, or comparator settings. The five 2021 reference dates end with the season-end measurement and only score estimates. Newly completed pole annotations describe 2021 images. Thus this is independent, label-held-out, subject-disjoint validation. The separate 2024-to-2025 test assesses year-forward uncertainty; every online update excludes future images.

In the simulation-only candidate extension, candidate j with detector evidence c_{tj} and implied height y_{tj} is scored before the measurement update as

$$j_t^* = \arg \max_j \{c_{tj} p(y_{tj} \mid \mathcal{I}_{\leq t-1})\}. \quad (S4)$$

This operation lets the height prior affect which top candidate becomes the measurement. It is distinct from applying the particle filter after an independently chosen detection, and it was not used

to generate the empirical field-height table.

S3 Manual 2021 support-pole annotations

Haoming Wang manually created the physical-reference annotations. The source annotation directory contains one CVAT XML file for each of 12 camera rows, named C-XXX.xml. An XML can contain multiple dated images, and each image can contain several polylines labeled pole1 through pole4. These labels distinguish visible camera-support poles within an image. The numeric suffix does not specify pole depth or physical length.

For each polyline, the importer reads all vertices, takes the smallest image row as the nominal camera-mount endpoint and the largest image row as ground contact, and records the sum of segment lengths in pixels. The physical definition is the nominal vertical distance from ground contact to the camera mounting or optical-center height: 5 ft (152.4 cm). Annotation status is read from the CVAT attribute named status. The imported data contain 199 traces: 172 marked usable and 27 marked unusable.

An earlier export contained two filenames for the same C-039 scene. Following the project decision, C-039_2021-07-30.JPG was retained and C-039_2021-07-30JPG.JPG was removed. The updated C-039.xml also includes LH54.21-C-039_2021-07-12-10-57_CAM827.JPG. The curated source therefore contains 61 unique XML-referenced images, and every referenced image was found in the source image directory.

The installation records separately specify four horizontal field-layout intervals of 35 in, totaling 140 in between adjacent rows. These are not graduations on the vertical support poles. They also differ from the 1-ft red-band spacing on the 2024–2025 calibration poles. The 2021 import is consequently an installation and annotation resource; it is not pooled with red-band calibration observations.

For the temporal repeatability audit, usable annotations were grouped by camera row and pole label. The 172 traces formed 39 series with at least two dated images across 12 rows. Within

each series, the sample standard deviation of polyline length was divided by its mean. The median coefficient of variation was 1.24% (95% whole-row bootstrap interval, 0.78–1.99%), and the 90th percentile was 3.65%. 38 of 39 series were below 5%; the maximum was 11.51%. The calculation uses all traces marked usable and is descriptive of projected-length stability. It neither removes the largest series nor treats temporal repeatability as proof of the nominal 5-ft length.

The importer writes three canonical products:

- `data/processed/manual_poles_2021/manual_pole_annotations_2021.csv`, one row per retained trace;
- `data/processed/manual_poles_2021/pole_definitions_2021.csv`, the physical definitions and source records; and
- `data/processed/manual_poles_2021/summary.json`, counts, duplicate decision, and image-presence audit.
- `outputs/pole_repeatability_2021/`, per-series coefficients of variation and the whole-row bootstrap summary.

S4 Physical calibration and depth correction

The 2024–2025 red-band pole supplies positions with known 30.48-cm spacing. Within one reference image, band image rows y_j are centered and scaled as $u_j = (y_j - \bar{y})/s_y$, and the calibration curve is

$$z(y) = \frac{au + b}{cu + 1}. \quad (\text{S5})$$

Only height differences are used. Consequently, an arbitrary additive physical datum along the pole does not affect a top-minus-base extent. Held-out bands evaluate interpolation, while the root-direction diagnostic removes lower bands, fits retained upper bands, and predicts downward. The reverse lower-to-upper calculation is not evidence about root extrapolation.

For the depth correction, place a camera at physical height H . Under the ideal pinhole model with negligible roll and a level shared ground plane, the image row of a point at height z and horizontal distance D can be written, up to the image-row sign convention, as

$$y(z, D) - y_{\text{hor}} = f_{\text{px}} \frac{H - z}{D}. \quad (\text{S6})$$

For two points separated by a vertical extent $h = z_{\text{top}} - z_{\text{base}}$, subtraction removes H and the horizon row:

$$|\Delta y(D)| = \frac{f_{\text{px}}}{D} h. \quad (\text{S7})$$

A scale learned at reference depth D_{ref} would assign a target pixel extent the value

$$h_{\text{pole scale}} = \frac{D_{\text{ref}}}{f_{\text{px}}} |\Delta y(D_{\text{tgt}})| = \frac{D_{\text{ref}}}{D_{\text{tgt}}} h_{\text{target}}. \quad (\text{S8})$$

Therefore,

$$h_{\text{target}} = \kappa h_{\text{pole scale}}, \quad \kappa = \frac{D_{\text{tgt}}}{D_{\text{ref}}}. \quad (\text{S9})$$

For the documented 2024–2025 geometry, $D_{\text{tgt}} = 8.5$ ft and $D_{\text{ref}} = 10.0$ ft, so $\kappa = 0.85$. The measurement standard deviation, being in the same linear units, is multiplied by the same factor. The factor is treated as fixed in the reported analysis; uncertainty in measured distances is not propagated.

Equation S9 is exact only within the model assumptions used in the derivation. Camera tilt or roll, lens distortion, uneven ground, different plant depths within a row, uncertain distances, and within-season camera movement can produce residual error.

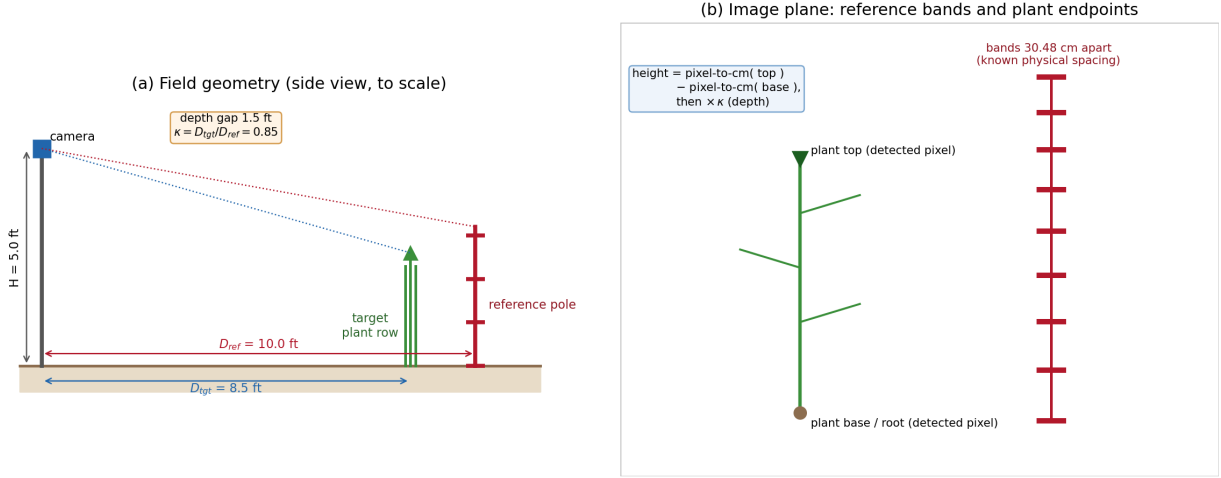


Figure S1: Idealized calibration-transfer geometry. (a) Side view with the 2024–2025 camera at 5 ft, the target row at $D_{\text{tgt}} = 8.5$ ft, and the red-band reference at $D_{\text{ref}} = 10.0$ ft, giving $\kappa = 0.85$. (b) Image-plane construction: the projective curve maps detected top and base rows onto the pole scale before the fixed depth factor is applied. The 30.48-cm labels refer to the red-band calibration pole, not the separate 2021 support poles.

S5 Height-growth particle filter and exact predictive mixture

For elapsed time Δ_t in days, the implemented transition order is

$$h_t^{(m)} = \text{clip}_{[0,450]} \left[h_{t-1}^{(m)} + \Delta_t r_{t-1}^{(m)} + \sigma_h \sqrt{\Delta_t} \epsilon_{h,t}^{(m)} \right], \quad (\text{S10})$$

$$r_t^{(m)} = \text{clip}_{[-4,15]} \left[T(h_t^{(m)}, r_{t-1}^{(m)}) + \sigma_r \sqrt{\Delta_t} \epsilon_{r,t}^{(m)} \right], \quad (\text{S11})$$

where the innovations are independent standard normal variables. For the daily pole stream, $T(h, r) = r$. For the 2021 transferred-growth-field configuration, $T(h, r) = \hat{r}(h)$, the nonparametric rate-versus-height field estimated from 2024–2025 image increments. Manual 2021 field heights do not enter $\hat{r}$.

Conditional on height particle $h_t^{(m)}$, the observation law is

$$p(Y_t = y \mid h_t^{(m)}) = (1 - \pi_t) \phi(y; h_t^{(m)}, \sigma_t^2) + \pi_t \phi(y; h_t^{(m)}, \sigma_{\text{out}}^2). \quad (\text{S12})$$

Let w_m^- be normalized particle weights after propagation and before observing Y_t . The exact one-step

predictive density and cumulative distribution are

$$p_t^-(y) = \sum_{m=1}^M w_m^- \left[(1 - \pi_t) \phi(y; h_t^{(m)}, \sigma_t^2) + \pi_t \phi(y; h_t^{(m)}, \sigma_{\text{out}}^2) \right], \quad (\text{S13})$$

$$F_t^-(y) = \sum_{m=1}^M w_m^- \left[(1 - \pi_t) \Phi\left(\frac{y - h_t^{(m)}}{\sigma_t}\right) + \pi_t \Phi\left(\frac{y - h_t^{(m)}}{\sigma_{\text{out}}}\right) \right]. \quad (\text{S14})$$

123 Predictive endpoints at probabilities 0.025, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95, and 0.975 solve
 124 $F_t^-(q_p) = p$ by numerical root finding. These quantiles provide central 50%, 80%, 90%, and 95%
 125 intervals. The predictive probability integral transform is $F_t^-(Y_t)$. The reported predictive standard
 126 deviation uses

$$\text{Var}(Y_t \mid \mathcal{I}_{\leq t-1}) = \sum_m w_m^- \left(h_t^{(m)} - \bar{h}_t^- \right)^2 + (1 - \pi_t) \sigma_t^2 + \pi_t \sigma_{\text{out}}^2. \quad (\text{S15})$$

127 Equations S14 and S15 retain both mixture components. They replace the earlier normal approxima-
 128 tion based only on latent-particle and inlier variances.

129 After Y_t is observed, the weights are updated as

$$w_m \propto w_m^- p(Y_t \mid h_t^{(m)}), \quad \sum_m w_m = 1. \quad (\text{S16})$$

130 The posterior mean and interval for latent height are computed from these updated weights. A
 131 posterior interval therefore targets h_t after the current image, whereas the predictive interval targets
 132 the next noisy image-derived measurement before it is observed. Coverage for those two targets is
 133 not interchangeable.

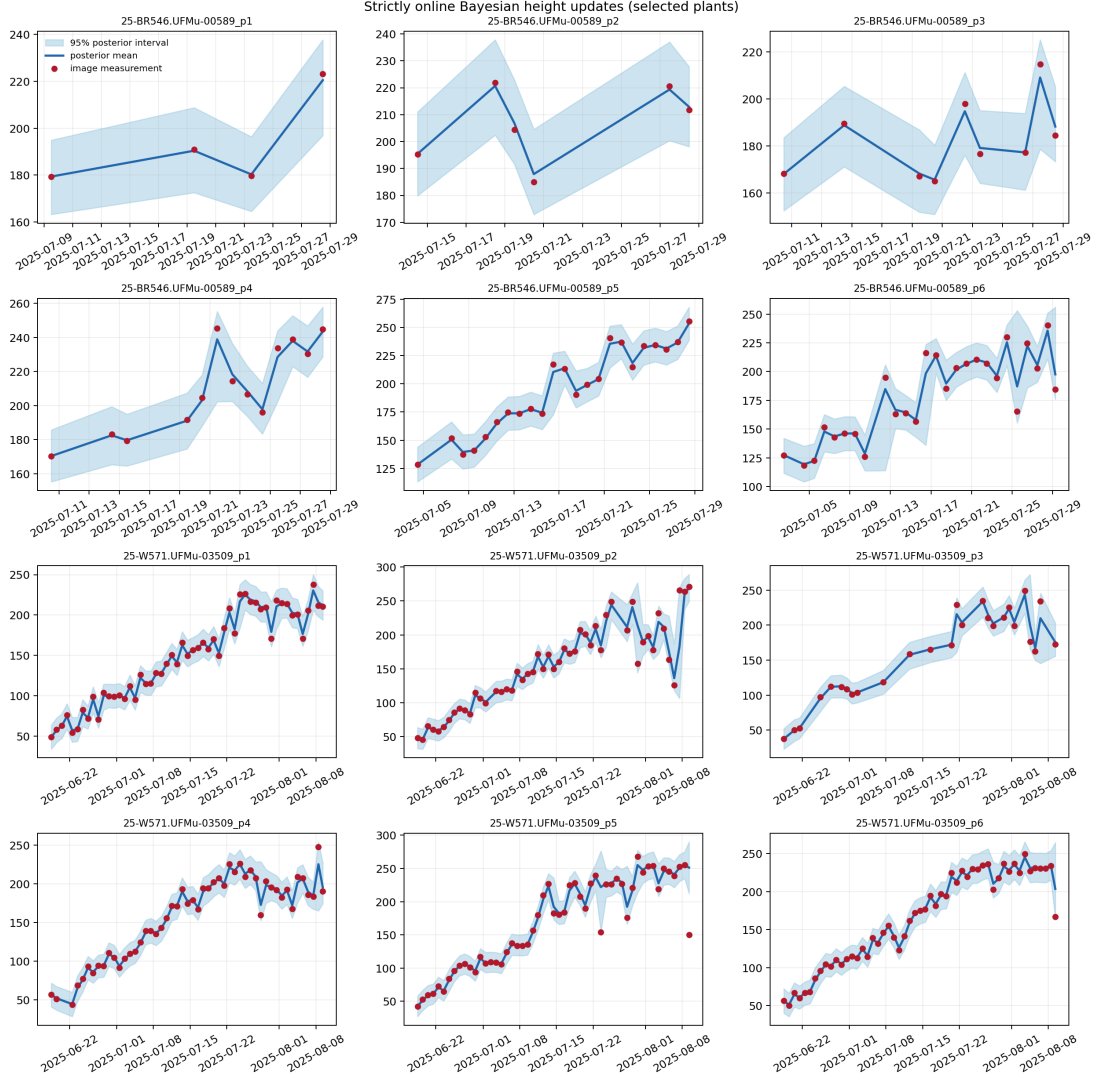


Figure S2: Selected 2025 daily pole-camera trajectories from the field-executed height-growth particle filter. Red points are depth-corrected image measurements, blue lines are posterior means after the corresponding image, and shading is the weighted 95% latent-height posterior interval. Later observations initiate later updates; they do not retrospectively revise the plotted earlier estimates. The figure illustrates posterior behavior against image-derived measurements and is not validation against manual plant height.

S6 Uncertainty against manual height and across years

This secondary analysis evaluates temporal transfer of predictive uncertainty for the next image-derived measurement. The 0.85 depth conversion changes every quantity expressed in centimeters. The corrected analysis therefore applies it to the image-derived height and its measurement standard deviation, and estimates the process and outlier scales from already corrected 2024 increments. The 2025 records are not used to estimate these values. Because plants were newly grown each season, the 2025 cohort is temporally later and subject-disjoint from 2024; no biological plant subject appears in both years. Over the complete quality-controlled stream, the unit-consistent predictive mean had mean absolute error 15.13 cm, mean log score -4.440, and weighted interval score 8.08 cm.

Table S2 and Figure S3 show calibration at multiple levels. On 372 year-forward 2025 forecasts from 11 evaluated plant tracks and 2 cameras, the 95% interval achieved 94.9% coverage with a 97.8-cm mean width. The 50%, 80%, and 90% intervals covered 65.1%, 86.3%, and 92.7%, respectively. This pattern is conservative at the lower levels even though the 95% result is near nominal. Subject disjointness prevents repeated plants across years; it does not make repeated forecasts within a plant or camera independent. With only two test cameras, this analysis supports temporal transfer for those installations and does not establish broad camera generalization.

The same table adds the primary 2021 physical-reference target. These intervals summarize latent height after the current image has been incorporated, and manual height is used only for evaluation. Central 80% intervals covered 87.1% of the 132 records with a 47.9-cm mean width; the 95% intervals covered 94.7% with a 75.4-cm mean width. Whole-camera-row bootstrap intervals were 79.3–93.6% and 89.4–98.6% for coverage, respectively. Per-date results and interval-width bootstrap summaries are released in `outputs/manual_height_uncertainty_2021/`.

Table S2: Multi-level uncertainty evaluation. The 2024–2025 rows evaluate one-step predictive intervals for the next image-derived measurement. The 2021 rows evaluate posterior intervals for latent height after the current image against held-out manual height.

Evaluation	Records	Clusters	Nominal (%)	Coverage (%)	Mean width (cm)
2024 development	1266	6	50	63.7	30.7
			80	87.8	59.4
			90	94.1	78.4
			95	96.8	98.2
2025 test	372	2	50	65.1	30.6
			80	86.3	59.3
			90	92.7	78.2
			95	94.9	97.8
2021 manual reference	132	12 rows	80	87.1	47.9
			95	94.7	75.4

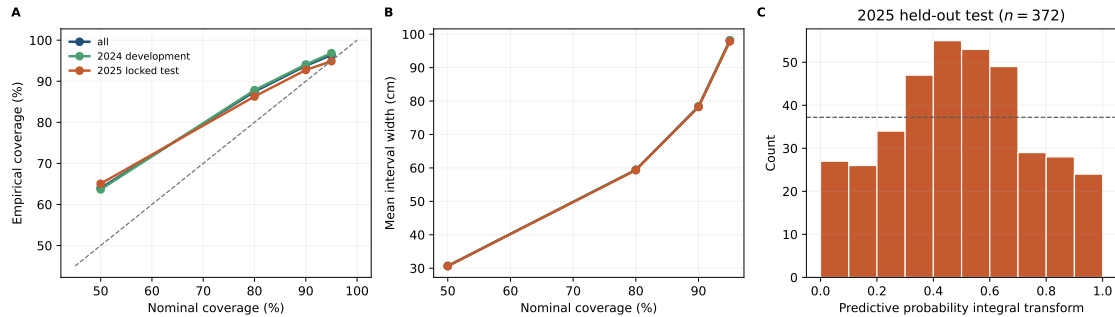


Figure S3: Secondary temporal evaluation of coverage and interval width for the unit-consistent predictive distribution. Parameters were estimated from the 2024 development cohort; the newly planted, subject-disjoint 2025 cohort was held out. The diagonal in the coverage panel marks exact nominal coverage. Points above the diagonal indicate conservative intervals.

S7 Matched real-image candidate audit

The real-image audit uses cached candidates from the released 2021 pose checkpoint. Each temporal method receives identical candidates, confidence scores, association gates, fixed scale (0.415886 cm/pixel), and particle-filter settings. Human-mask root positions initialize the first image only. Subsequent associations use the running state; manual field height is reserved for scoring. The matched subset contains 63 plant-dates from 17 tracks in 6 cameras.

The post-detection and prior-in-candidate methods agreed on every selected candidate (Table S3). Their errors are consequently identical. This is a useful negative control: under these natural-image candidates and gates, there was no ambiguity for the height distribution to resolve. It cannot verify the accuracy gain seen in the controlled ambiguity simulation.

Table S3: Real-image candidate audit. Manual field heights were used only for scoring.

Method	Signed error (cm)	MAE (cm)	RMSE (cm)
Single frame	-9.55	13.38	18.01
Identical candidate plus post-detection filter	-9.27	13.07	17.72
Height prior in candidate scoring plus same filter	-9.27	13.07	17.72

S8 Field-error distribution and row influence

The primary comparison uses record-weighted mean absolute error and its whole-row bootstrap interval. As a secondary robustness audit, Table S4 reports a complete set of absolute-error quantiles for every fixed causal estimator. The robust Bayesian particle filter had the lowest error at each reported quantile. Its median absolute error was 7.30 cm, compared with 8.74 cm for the single-frame extent. The paired 1.43-cm median reduction had a 0.29 to 3.70 cm whole-row bootstrap interval.

Influence was evaluated by removing each stationary-camera row in turn and recomputing the record-weighted mean absolute-error difference. All 12 leave-one-row-out estimates favored the particle filter, ranging from 0.61 to 1.42 cm. These checks show that the direction of the observed

176 advantage is not created by one camera row or only by extreme errors. They remain secondary to
 177 the prespecified mean absolute-error comparison, whose cluster interval includes zero.

Table S4: Secondary absolute-error distribution for the independent 2021 manual-reference validation. Values are centimeters across all 132 plant-date records; lower values are better.

Estimator	25th	Median	75th	90th	95th
Single-frame image extent	3.35	8.74	15.32	26.01	31.51
EWMA	5.38	11.92	21.08	29.87	37.25
Running median	5.37	12.13	25.26	42.97	65.61
Holt level–trend	5.24	9.85	20.34	28.57	35.30
Gaussian local-linear	3.40	8.41	15.56	25.94	31.76
Kalman					
Robust Bayesian particle filter	3.31	7.30	15.15	23.35	30.35

S9 All-plant mature-date results

Table S5 lists the last available 2021 observation for every plant retained in the primary independent manual-reference validation. No plant is removed using image-minus-field residuals. The single-frame column is the archived geometrically scaled mask extent; the Bayesian column is the height-growth particle-filter posterior mean. The interval is the 2.5th to 97.5th percentile of the latent-height posterior after the endpoint image. Manual endpoint height is used only to score the corresponding prediction and is not supplied to the filter.

Table S5: All 33 plants at their last available 2021 field date.

Heights and interval endpoints are in centimeters.

Plant	Manual field	Single frame	Bayesian mean	Bayesian 95% interval
C_004-p1	110.0	110.8	109.2	[79.2, 142.4]
C_004-p3	132.0	129.8	149.0	[120.3, 178.9]
C_004-p4	140.0	144.5	138.0	[108.4, 169.2]
C_005-p1	144.0	153.9	128.3	[98.5, 158.7]
C_005-p4	130.0	125.6	136.4	[106.8, 167.3]
C_005-p5	138.0	133.9	136.6	[106.7, 167.7]
C_009-p1	148.0	146.8	151.1	[121.6, 182.2]
C_009-p2	132.0	133.5	140.1	[111.9, 170.8]
C_009-p3	136.0	146.0	154.7	[124.6, 185.4]
C_010-p1	116.0	116.0	123.2	[92.1, 157.1]
C_010-p2	112.0	120.8	127.0	[95.9, 161.0]
C_010-p3	70.0	98.1	80.3	[28.5, 136.8]
C_011-p1	196.0	211.9	204.0	[141.2, 239.5]
C_011-p2	196.0	210.4	205.6	[137.7, 244.5]

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Table S5 continued

Plant	Manual field	Single frame	Bayesian mean	Bayesian 95% interval
C_011-p3	196.0	216.7	198.6	[149.6, 231.2]
C_021-p3	48.0	75.9	45.3	[11.6, 79.1]
C_021-p4	110.0	86.9	117.8	[86.8, 159.0]
C_024-p1	116.0	137.2	122.6	[91.1, 157.4]
C_024-p2	112.0	122.7	130.2	[99.5, 164.1]
C_024-p3	118.0	123.9	125.0	[94.8, 158.8]
C_029-p1	210.0	212.1	208.7	[178.5, 238.0]
C_029-p4	200.0	217.9	216.9	[185.9, 246.8]
C_029-p5	190.0	216.1	212.8	[182.7, 242.6]
C_032-p1	130.0	124.6	125.2	[93.8, 158.6]
C_032-p2	120.0	136.4	138.4	[107.2, 170.6]
C_032-p4	140.0	142.6	129.7	[99.2, 164.6]
C_037-p2	90.0	91.9	113.4	[83.5, 145.5]
C_037-p3	100.0	98.1	117.4	[88.8, 148.8]
C_037-p4	112.0	115.8	110.6	[80.6, 144.4]
C_038-p1	142.0	84.8	83.2	[51.9, 116.8]
C_039-p1	120.0	154.7	147.1	[117.6, 176.6]
C_039-p2	110.0	155.1	144.4	[113.8, 176.6]
C_039-p3	140.0	147.2	136.7	[107.0, 167.5]

The all-record analysis includes first images so that every available manual reference remains in the primary estimand. A secondary analysis isolates the 99 records after a plant's first image, when longitudinal information from a preceding image exists. Across these records, single-frame and Bayesian mean absolute errors were 12.58 and 11.14 cm. The 1.44-cm reduction had a whole-row

bootstrap interval of -0.21 to 3.26 cm. This outcome-independent restriction preserves the direction of the primary estimate but remains statistically imprecise.

Across these 33 endpoints, single-frame mean absolute error was 14.45 cm and the Bayesian mean absolute error was 12.38 cm. The Bayesian signed error was +6.17 cm. The stationary-camera-row bootstrap interval for the 2.08-cm within-plant mean-absolute-error reduction was -1.47 to 6.82 cm. These endpoint results are secondary because the interval is wide and includes zero; the main article uses all 132 plant-date records as its primary comparison.

S10 Data provenance and reproducibility

All paths below are relative to the public GitHub repository.

Table S6: Data and result provenance for the revised submission.

Asset	Canonical path or source	Role and limitation
2021 pole XML	data/manual_annotations/poles_2021/C-*.xml	Haoming Wang’s human physical-reference annotations; one XML per camera row
2021 pole images	data/raw/pole_calibration_images/	All 61 curated images; confirmed duplicate omitted
Imported 2021 pole table	data/processed/manual_poles_2021/manual_pole_annotations_2021.csv	Trace endpoints, status, pixel length, and physical definition after duplicate removal
Pole temporal repeatability	outputs/pole_repeatability_2021/	All 39 repeated usable row–pole series; within-series pixel-length CV and whole-row bootstrap summary
Pole definitions	data/processed/manual_poles_2021/pole_definitions_2021.csv	Documents 5-ft support length and distinguishes horizontal 35-in marks from red-band spacing
Daily image manifest	data/processed/pole_camera_daily_manifest.csv	As-of capture records, local/remote identifiers, and checksums
Released pose checkpoint	models/maize_pose_2021_best.pt	2021 YOLO pose weights with SHA-256 checksum; embedded software version, seed, deterministic setting, training arguments, and metrics are recorded in models/checkpoint_metadata.json, while the referenced dataset file, exact image split, original masks, and source commit remain unavailable

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Table S6 continued

Asset	Canonical path or source	Role and limitation
Curated-image pose candidates	outputs/pose_candidates_annotation_set_2021/pose_candidates.csv	Complete cache for all 61 released images
Automatic calibrations	data/processed/automatic_pole_calibrations_daily.csv	Candidate red-band fits and quality-control fields
Causal audit	outputs/online_study/causality_audit_daily.json	Prefix recomputation for cached detections through measurement assembly
Daily Bayesian output	outputs/online_study/pole_camera_bayesian_daily_revised/online_bayesian_height_posteriors.csv	Unit-consistent exact-mixture predictive and posterior summaries
Daily model settings	outputs/online_study/pole_camera_bayesian_daily_revised/model_hyperparameters.json	Complete saved particle-filter settings
2021 primary independent validation	data/raw/heights_compare_2021.csv	Archived image extents and 132 manual field measurements across 12 stationary-camera rows; 63 records across six rows have recovered source-frame coordinates, while 69 retain only the derived extent
Partial source-frame recovery	outputs/crop_positions_2021/crop_positions.csv	Coordinates and masked-template match scores for 87 archived crop records across seven camera rows; 63 records join the primary validation, and source pixels are not redistributed in this release
2021 canonical posterior	outputs/filter_height_sam_2021/filtered_height_sam__phi1.0_plus_pole_growth_field.csv	Input to all-plants and per-plant summaries
Primary field summary	outputs/filter_height_sam_2021/all_plants_primary.json	All-record and endpoint metrics with stationary-camera-row-cluster intervals; regenerate with <code>python reproduce_core.py</code>
Manual-height interval audit	outputs/manual_height_uncertainty_2021/	Central 80% and 95% posterior coverage, width, per-date summaries, and whole-row bootstrap intervals
Detector-unit correction	outputs/pixel_validation_2021/summary.json	Recomputed centimeter metrics using 0.41588595 cm/pixel rather than the withdrawn field-regression conversion

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Table S6 continued

Asset	Canonical path or source	Role and limitation
Field baselines	outputs/field_baselines_2021/	Fixed causal comparators, absolute-error quantiles, row-influence audit, and row-cluster intervals
Real-image candidate audit	outputs/real_image_candidate_ablation_2021/	Matched candidate comparison with human truth used only for scoring